Introducing AI to an Online Petition Platform Changed Outputs but not Outcomes

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The rapid integration of AI writing tools into online platforms raises critical questions about their impact on content production and outcomes. New AI writing tools and assistants based on Large Language Models (LLMs) are capable of generating high-quality text that is more articulate, persuasive, and creative than if the users had written the text themselves [6, 3, 5, 1, 4]. However, the effect of in-platform AI writing tools on content and its outcomes on online platforms remains uncertain, with conflicting evidence regarding the contexts in which AI can augment human performance [8]. While generally helpful, the effect of AI assistance on text performance could be task-specific; AI might not be as effective in tasks that are more nuanced.

We leverage a unique natural experiment on Change.org, a leading social advocacy platform, to causally investigate the effects of an in-platform "write with AI" tool. In 2023, Change.org integrated an AI writing tool built on an OpenAI-provided LLM into the petition writing workflow. The site's home page encouraged visitors to "use AI to create a compelling petition in minutes." Petition writers can use this tool to generate petition drafts from only a short, user-entered description of a cause. Petition outcomes, and thereby the outcomes of the causes they champion, may be influenced by AI tools. Conceivably, integration of AI could bolster the effectiveness of the Change.org platform, as well as individual petitions, by lowering the barrier to participation and enabling users to produce high quality content with ease.

To understand the impact of the AI integration, we collected 1.5 million petitions written between 2022 and 2024, and used a period of differential AI tool access to employ a difference-in-differences analysis of the tool's impact. Our findings reveal that the introduction of the AI tool significantly altered the lexical features of petitions on the platform. However, this shift came at a cost: access to the AI tool significantly increased the homogeneity of petitions. Crucially, access to the AI tool did not improve—and possibly worsened—petition outcomes, as petitions were not more likely to succeed in hitting minimal thresholds of comments or signatures. The results suggest that while AI writing tools can profoundly reshape online content, their practical utility for improving desired outcomes may be less beneficial than anticipated, while potentially introducing unintended consequences like content homogenization.

Methods In this work, we leveraged a unique natural experiment to estimate the causal impact of access to the Change.org AI writing tool on published petitions in terms of lexical features and petition outcomes. Historically, the user experience on Change.org had been similar in the United States, Great Britain, Canada and Australia. However, this newly introduced AI tool was first rolled out in the United States, Great Britain, and Canada, with a delayed release in Australia. We conducted a difference-in-differences analysis that leverages this staggered roll-out in AI tool access to measure the causal impact. Unlike off-platform AI tools, the Change.org in-platform AI tool is uniquely tailored to the task and thereby channels AI usage through a single model, a single backend prompt, and standard interface. This in-platform AI tool and its staggered roll-out allowed us to avoid concerns that arise in many observational studies of generative AI use in the wild, where AI use can be difficult to detect and driven by idiosyncratic user flows.

On Change.org, the "write-with-AI" tool was released ten weeks earlier in the United States, Great Britain, and Canada than in Australia. We defined the intervention in our causal analysis as AI access, rather than the actual use of AI in the petition. Since all countries satisfy parallel trends in petition style and outcomes prior to AI tool release, we can attribute differences in trends during the ten-week period to the introduction of the AI tool. Furthermore, after the release of the AI tool in all geographic regions, we observe parallel trends resume. Our comparison of trends followed a difference-in-differences quasi-experimental approach, in which we measured the change in trends between similar groups after some groups received an intervention. We leverage two types of specifications for the difference-in-differences approach: static and dynamic. In the static difference-in-differences specification, we measure the overall effect across countries and weeks of AI access for treated countries. In the dynamic difference-in-differences specification, we measure the effect of AI access for each week in which treated countries have AI access [2].

Empirical Results

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Lexical Features We estimated the effect of in-platform AI tool access on petition lexical style through three features: lexical diversity, readability, and text length. Lexical diversity and readability have been used to assess writing competency in a variety of contexts such as language proficiency and clarity, and sufficient length is necessary for a petition to convey a compelling message. We operationalized lexical diversity through Moving Average Type-Token Ratio (MATTR), a measure of lexical diversity calculated as a moving average of the ratio of unique words to total words across windows of the text. Higher MATTR scores denote a more varied vocabulary. We operationalized readability using the Flesch-Kincaid grade level, a common measure of readability calculated as the number of words per sentence and syllables per word. Higher Flesch-Kincaid grade levels denote more complex text. For text length we used the word count, namely the number of words in the petition text and title.

Figure 1a demonstrates that **access to the AI tool had a significant and consistent effect on lexical features**: petitions were longer, with more complicated and varied vocabularies. Figure 1a shows the results of a dynamic difference-in-differences estimation, in which we estimate the average treatment effect in treated countries by week (x-axis), compared to the baseline (week -1). Weeks in orange are before any country had access to the AI tool (up to week -1), gray are the A/B period in treated countries (week 0 through week 25) and blue are after the full launch in treated countries (week 35). The shift is captured via average group-time treatment effect estimates on treated countries, relative to the baseline (week -1).

Figure 1b shows the result of a static difference-in-differences estimation, by which the average treatment effect is estimated across the post-AI (after week 25) period, relative to the pre-AI period (before week 0). We exclude the A/B testing period from the post-AI analysis because our data does not indicate which users had AI access during A/B testing. Median MATTR (shown in Figure 1b) increased by 0.049 units (95% CI = [0.048, 0.050]), median petition grade level (second row of Figure 1b) increased by 1.51 grades (95% CI = [1.22, 1.79]), and median word count (bottom row of Figure 1b) increased by 55.15 words (95% CI = [48.63, 61.68]). An increase of 50 words corresponds to a 45% increase in median petition length, from 110 words; the median FKGL increased by 19%, from grade 8; and the MATTR increased by 6%, from 0.8 units. A descriptive analysis shows that petition vocabularies qualitatively changed as well: prior to AI access the most frequent verbs in petition titles were mono-syllabic Old English verbs such as "stop" and "let". Following the AI tool's release, multi-syllabic verbs like "implement" and "establish" jumped in frequency.

Platform Homogeneity We also find **access to the AI tool increased in inter-petition homogeneity**, as measured through the mean pairwise similarity between petitions [7]. To measure homogeneity, we first created 3,072-dimension embeddings with the OpenAI text-embedding-3-large model[1] for all petitions. We then calculated the cosine similarity between all pairs of petitions posted in a given week and country, and took the mean of these pairwise scores to get the mean pairwise similarity for that week and country. We used these weekly country-level time series of homogeneity to compare trends in treated and untreated countries through a dynamic difference-in-differences estimation. The dynamic difference-in-differences results shown in Figure 2 demonstrate that access to the AI tool caused petitions to be more similar to one another than if they had been written without access to the AI tool. In weeks after week 25, following the introduction of AI, AI access had a consistent effect on petition homogeneity, increasing mean pairwise similarity by about 0.05 units (on a scale of -1 to 1) for each period.

Outcomes We find that **access to the AI tool did not improve petitions' outcomes**—and in some cases outcomes worsened. We examined petition outcomes as the share of petitions that reach 1 comment in 30 days and the share that reach a total of 10 signatures at the time of data collection. As each petition was posted at least one year prior to data collection, we expect recency was not a factor in whether petitions reached these minimal thresholds. We use these minimal thresholds instead of raw engagement counts because engagement is sparse, and petitions on Change.org have varied scale from hyper-local neighborhood issues to global issues. By using minimum thresholds we ensure our analysis is not biased towards issues that affect more populous communities while still reflecting a meaningful indicator of a petition's outcomes.

We do not find evidence that AI tool access positively impacted petition outcomes. Instead, we observe that with AI access, outcomes are either unchanged, with no significant difference from 0, or slightly worsened. Figure 1c shows the results of a dynamic difference-in-differences estimation, in which we find that the average estimated change in treated countries' comments and signatures tends to be worse than the baseline week (Week -1) during the weeks post-AI access. For example, the top graph in Figure 1c shows that in the

[1] `https://platform.openai.com/docs/models/text-embedding-3-large`

weeks after the full launch (first row, blue, right of Week 25), the share of petitions that received at least one comment within 30 days is consistently lower than the baseline week. Figure 1d shows that after AI tool release, the share of petitions that receive a comment in 30 days decreased by -1.11 percentage points relative to the share of petitions pre-AI tool release, although that difference was not statistically significant (95% CI = [-4.12, 1.90]). We also observe that the share of petitions that reached 10 signatures when written with access to AI decreased by -4.65 percentage points, which was statistically significant (95% CI = [-7.74, -1.55]). In sum, we do not find evidence that petition outcomes are improved when users have access to AI; indeed, some evidence suggests they may worsen.

Participation We did not find evidence that the effect of AI access on platform trends was driven by changed participation patterns: participation on Change.org did not increase with access to the AI tool. Furthermore, we demonstrate that changes to platform composition were not responsible for changes to platform trends by showing that the platform-level trends in outcomes are consistent even when examining trends within repeat users. We focus our analysis here on one outcome (comment engagement). We do not conduct a panel difference-in-differences analysis because of the rarity of repeat petition writing; only 7.6% of users in our dataset wrote multiple petitions, which amounted to too few untreated users for a reliable balanced panel. Instead, we model the predictive relationship between AI access and word count or comment engagement among repeat writers from treated countries (the United States, Great Britain, and Canada) over the full period for which we collected data.

We found that, across cohorts, returning writers' second petitions had worse outcomes than their first petitions, and that this decline was greater when users had AI access for their second petition. We use our preregistered primary outcome metric, the presence of at least one comment in 30 days threshold, as an indicator of petition outcomes. In the Pre/Post cohort (middle facet of Figure 3), 30% (95% CI = [28.35, 31.91]) of writers' first petitions (first bar, orange) and 25% (95% CI = [23.77, 26.91]) of writers' second petitions (second bar, blue) received at least one comment in 30 days. We estimated the predictive relationship of AI access on comment engagement by fitting a mixed-effects logistic regression model. Through Bonferroni-adjusted pairwise contrasts, we find that Pre/Post users' second petitions, which were written with access to AI, have 34% lower odds of a positive outcome than their first petitions, with this difference being highly significant (OR=0.709, z=-4.61, $p < 0.001$). On the other hand, the first and second petitions written by users in the Pre/Pre cohort, for whom both petitions were written without AI access, have no significant difference in outcomes (OR=0.871, z=-1.23, p=0.652). In the Post/Post cohort we observe a similar pattern to the Pre/Post cohort, with a significant difference in odds of first and second petition positive outcome (OR=0.595, z=-3.851, $p < 0.001$). The decline in petition outcomes for petition writers' second petitions across cohorts reflects a predictable regression to the mean. However, we also see that among repeat petition writers, AI access is associated with significantly worse petition outcomes. The negative relationship between AI access and petition outcomes among repeat-petition writers extends our findings from the platform-level difference-in-differences analysis, in which we see no evidence that AI access improved petition outcomes.

Conclusion Our work provides a causal analysis of the impact of AI integration into a real-world online platform. We affirm our findings with synthetic control analysis and a series of robustness checks (controlling for potential confounders, using alternative lexical features and outcome metrics, using log count of outcomes instead of minimal thresholds, including A/B testing period in differential period) and placebo tests. Our findings extend prior work on the impact of AI on online platforms: while online platforms might be influenced by both off-platform AI and in-platform AI tools, past studies have focused on off-platform AI. As online platforms and services increasingly integrate AI into their user flow, it is important to understand the potential impact of in-platform AI on platform dynamics. We find that in-platform AI on Change.org shifts petition lexical features and increases site-wide homogeneity. At the same time, the in-platform AI tool did not improve platform outcomes—potentially making them even worse—nor did it seem to increase platform participation. We have three hypotheses for the lack of change in outcomes: first, AI might not substantively improve content quality as perceived by readers; second, AI might increase readers' AI suspicion; and third, AI-assisted writing might reduce petition-writer ownership and effort. With AI minimizing the cognitive load of petition-writing, petition writers might not only feel diminished ownership but also less engagement with the petition's cause and their community at large. In sum, AI integration in petition-writing might significantly change the dynamics of civic engagement in ways that are difficult to predict and measure. Overall, our study serves as a cautionary tale for platforms that seek to integrate AI into their platforms:

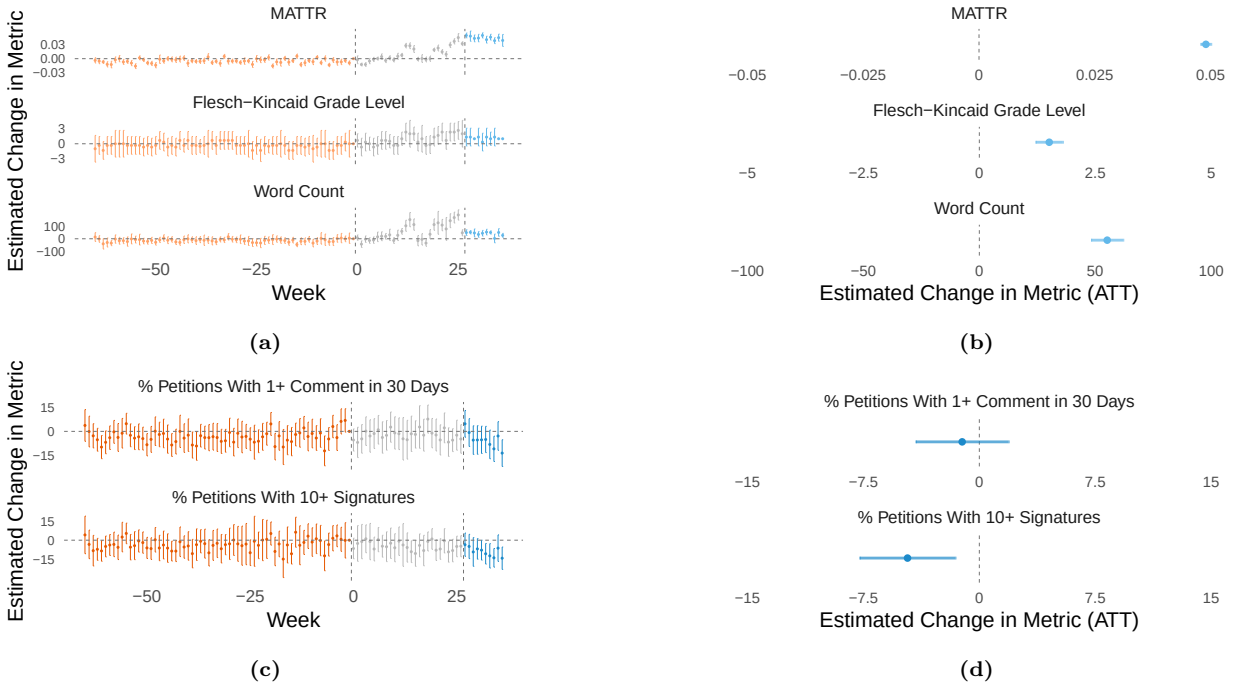

Figure 1: Difference-in-differences (DiD) analysis of platform trends in petitions' (N = 277,496) lexical features (a, b) and outcomes (c,d) before and after Change.org AI tool release shows that lexical features change significantly while outcomes do not improve, or worsen. Change in outcomes reflect change in percentage points. (a) Dynamic difference-in-differences with estimated effect of AI access on lexical features by week, relative to the baseline week (Week -1). (b) Aggregated estimated effect of AI access on lexical features, relative to pre-AI period. (c) Dynamic difference-in-differences with estimated effect of AI access on outcomes by week. (d) Aggregated estimated effect of AI access on outcomes, relative to pre-AI period. The A/B testing period, shown in gray between the dashed lines, is excluded from analysis. 95% confidence intervals are shown.

while user-generated content may change, AI tools may not yield dividends in participation or effectiveness, while potentially introducing additional harms. We urge the research community to continue to develop research that seeks to better understand and ameliorate these emergent harms.

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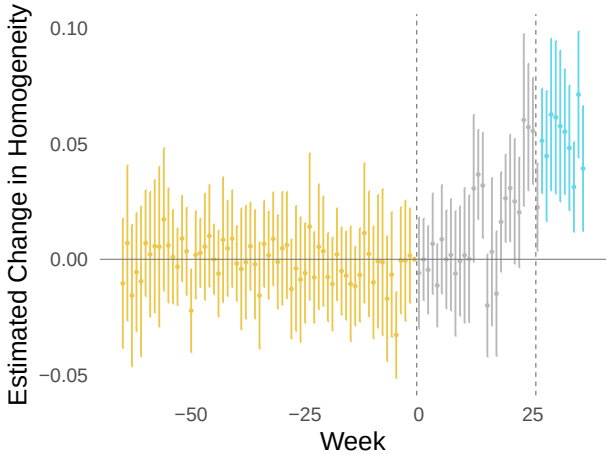

Figure 2: Dynamic difference-in-difference estimates for homogeneity in countries with access to AI shows that with access to the AI tool, platform homogeneity increases (N = 277,496). Estimated average treatment effect (ATT) of mean pairwise similarity is shown with 95% CIs. Estimates are compared to baseline of the week prior to A/B testing in treated countries (Week -1).

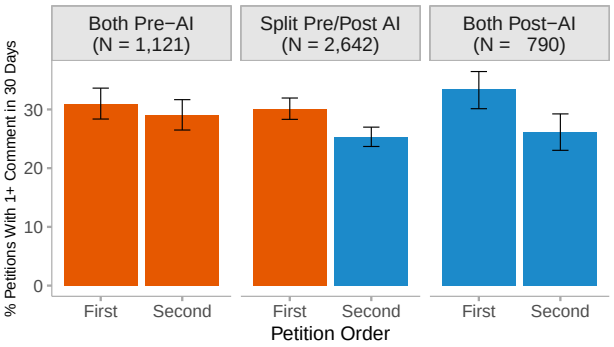

Figure 3: Returning petition writers (N=4,553 users) from treated countries, on average, achieve worse outcomes when their second petition is written with access to AI. Petition outcomes are worse in writers' second petitions, in all conditions. The greatest drop in outcomes for users' second petitions occurs when the second petition is written with access to AI. 95% CIs (bootstrapped, 1,000 iterations) are shown. Returning petition writers who return within 6 months are excluded, to enforce the same minimum time gap between petitions across cohorts.